

EXPERIMENT 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory

management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js.

html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Inventory Management Visualization</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
text-align: center;
```

```
margin: 50px;
```

```
}
```

```
canvas {
```

```
margin: 20px auto;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Inventory Management System</h1>
```

```
<canvas id="pieChart" width="400" height="400"></canvas>
```

```
<canvas id="barChart" width="400" height="400"></canvas>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic.

javascript

script.js

```
// Data for the inventory
```

```
const inventoryData = {
```

```
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
```

```
  datasets: [
```

```
    {
```

```
      label: 'Items in Stock',
```

```
      data: [200, 150, 100, 80, 50],
```

```
      backgroundColor: [
```

```
        '#FF6384',
```

```
'#36A2EB',
```

```
'#FFCE56',
```

```
'#4BC0C0',
```

```
//
```

```
'#9966FF'
```

```
],
```

```
}
```

```
]
```

```
};
```

```
// Creating the Pie Chart
```

```
const ctxPie = document.getElementById('pieChart').getContext('2d');
```

```
const pieChart = new Chart(ctxPie, {
```

```
  type: 'pie',
```

```
  data: inventoryData,
```

```
  options: {
```

```
    responsive: true,
```

```
    title: {
```

```
      display: true,
```

```
      text: 'Inventory Distribution'
```

```
    }
```

```
  }
```

```
});
```

```
// Creating the Bar Chart
```

```
const ctxBar = document.getElementById('barChart').getContext('2d');
const barChart = new Chart(ctxBar, {
  type: 'bar',
  data: inventoryData,
  options: {
    responsive: true,
    title: {
      display: true,

      text: 'Items in Stock by Category'
    },
    scales: {
      yAxes: [{
        ticks: {
          beginAtZero: true
        }
      }]
    }
  }
});
```

Inventory Management System

Electronics Clothing Home Appliances
Books Toys

